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Loan Application Proposal	
Client	Mr. Imran
Vendor/Service Provider	Comfygen Technologies
Prepared by	Bhavika Gupta
Date	11 Dec, 2025
Version	1.0
Estimated Cost	₹ (approximately)
Estimated Timeline	Months – Working Days (approximately)

Important Note

Project timeline and cost will commence only after formal confirmation and approval from the client that the documented features and requirements fully meet their expectations.

Proposal Disclaimer

This document serves as an **initial project proposal** based on preliminary analysis and discussions. It outlines the **scope, technical architecture, and estimated costs** but does **not** serve as the **Functional Requirement Document (FRD)**.

Upon signing the contract, we will conduct a **detailed requirement analysis** with the client and draft an **in-depth FRD** that will include:

- **Comprehensive feature definitions.**
- **Detailed UI/UX wireframes for each screen.**
- **Specific user flows and system interactions.**
- **Finalized scope of work with precise functionalities.**

If any **feature, module, or requirement** is missing in this proposal, it will be incorporated into the **FRD** after a thorough requirement gathering phase. The final implementation will be aligned with the client's business needs, ensuring **complete functionality, scalability, and user experience enhancements**.

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1. Executive Summary

The proposed Loan Origination Platform is a comprehensive digital solution designed to streamline and modernize agent-driven lending operations across Tier 1, Tier 2, and small-city markets. This platform empowers field agents to digitally onboard customers, conduct KYC verification, perform CIBIL and CRIF credit checks, collect required documents, capture non-refundable application fees, and submit loan applications with accuracy and speed. The Admin Web Portal complements the agent app by offering full operational control, underwriting capabilities, loan approval workflows, and compliance-ready audit systems.

The solution eliminates traditional paper-based, manual, and error-prone processes by introducing a structured, digital-first workflow that improves turnaround time and enhances data integrity. A key USP of the system is its **agent-centric design**, enabling agents to serve customers even in remote areas using a mobile app optimized for low bandwidth, manual document uploads (without OCR), offline usage, and seamless synchronization.

Once an application is reviewed and approved, the system automatically generates a **Consent/Loan Sanction Letter**, which is shared with the customer via email or WhatsApp. The agent then meets the customer physically to collect bank account details, disbursal forms, and SLA documentation—ensuring a compliant and authenticated disbursal process. If a loan is rejected, the system sends a clear rejection notification to the customer.

The platform also includes an EMI calculator for agents, automated customer reminders 2–3 days before due dates, offline non-refundable fee handling, and conversion of **Applicant ID into a Loan Account Number** once the loan is approved. These features collectively provide transparency, efficiency, and consistency across the entire loan lifecycle.

By adopting this solution, the client can expect significant business value, including:

- **Reduced processing time & improved accuracy** through structured, guided workflows.
- **Higher agent productivity** in the field with offline capabilities and simplified data entry.
- **Better credit decisioning** with integrated CIBIL/CRIF score checks and underwriting tools.
- **Enhanced customer experience** through quicker communication, status visibility, and timely notifications.
- **Stronger compliance and audit readiness**, including digital consent records, versioned documents, and a robust approval trail.
- **Scalability for growth** as lending expands across multiple geographies and customer segments.

Ultimately, this platform positions the client as a modern, technology-enabled lender capable of delivering **faster, safer, and more transparent credit services**, while maintaining strict regulatory compliance and operational efficiency. It creates a strong foundation for future enhancements such as online EMI payments, automated disbursal, NACH/UPI mandate collections, and a customer-facing portal.

2. Introduction

2.1 Background / Problem Statement

Many financial institutions, especially those operating across Tier 1, Tier 2, and small cities, continue to rely heavily on manual, paper-based loan onboarding processes. Field agents collect customer information, KYC documents, and income proofs physically, leading to operational inefficiencies, long turnaround times, missing documents, and compliance risks. Manual verification of documents and offline fee collection add to financial leakage and administrative burden.

Additionally, lenders face challenges such as:

- **Inconsistent data quality and incomplete applications**, causing delays.
- **Scattered communication** between agents and back-office teams.
- **Limited visibility** into real-time application progress.
- **Difficulty verifying identity and creditworthiness**, especially for first-time borrowers.
- **Dependence on physical meetings** for consent, disbursal forms, and bank account collection.
- **Operational friction** in small and remote regions with low connectivity.
- **Lack of structured customer communication**, including rejection notices and EMI reminders.

These challenges restrict scalability, reduce agent productivity, and increase the risk of regulatory non-compliance. A unified digital system is required to streamline the entire lifecycle—from agent onboarding to loan approval, consent, and repayment reminders—while retaining the field-based workflow needed in smaller markets.

2.2 Objectives of the Project

The primary objective is to build a **complete digital Loan Origination Platform** that enhances operational efficiency, ensures compliance, and improves customer experience—while remaining practical for on-ground agent workflows.

Core Objectives

1. **Digitize the Loan Application Process**
   Enable agents to capture customer details, upload KYC documents, record fees, and submit applications digitally—without relying on physical paperwork.
2. **Strengthen KYC & Credit Verification**
   Integrate Aadhaar/PAN verification and CIBIL/CRIF checks to improve decision accuracy and reduce risk.
3. **Streamline Underwriting & Loan Approvals**
   Provide admins with a centralized review portal to verify documents, evaluate credit results, and approve or reject applications efficiently.
4. **Enable Customer Consent & Disbursal Compliance**
   Automate the generation of consent/sanction letters, facilitate physical customer meetings, and support upload of disbursal forms, bank details, and SLA documentation.

5. **Improve Customer Communication**
   Send automated notifications including Applicant ID, sanction letters, rejection messages, and EMI reminders before due dates.
6. **Support Offline Fee Collection & Reconciliation**
   Allow agents to record cash/branch deposits while enabling finance teams to reconcile fees against bank statements.
7. **Ensure Scalability, Security & Compliance**
   Implement secure data handling, audit trails, encryption, and regulatory compliance, ensuring readiness for future expansion.

2.3 Proposed Solution Overview

The platform consists of two core components, powered by a centralized backend and seamless integrations.

A. Agent Mobile Application (Android & iOS)

A field agent–focused app that allows agents to:

- Register/login securely (OTP or biometrics).
- Capture customer information and upload required documents manually (no OCR).
- Perform Aadhaar/PAN KYC verification and trigger CIBIL/CRIF credit checks.
- Record non-refundable application fees (cash/branch deposit).
- Submit loan applications and receive an **Applicant ID** instantly.
- Receive notifications and respond to admin requests for more documents.
- Review application status and updates.
- Access a built-in EMI calculator.
- After approval, meet customers to collect bank account details, signed disbursal forms, and SLA documentation, then upload them digitally.

This mobile app works in both online and offline environments, ensuring agents in remote locations can operate without disruption.

B. Admin Web Portal

A powerful web-based system for internal teams enabling:

- Review of all submitted applications and customer documents.
- Manual verification of identity, income, and supporting files.
- Viewing credit scores from CIBIL/CRIF and generating underwriting notes.
- Approving or rejecting applications, with reasons and audit logs.
- Automatic generation and customer delivery of **Loan Sanction Letters**.
- Conversion of **Applicant ID into a Loan Account Number** upon approval.
- Tracking and validating uploaded bank details, disbursal forms, and SLA documentation.
- Managing agent accounts and performance metrics.
- Reconciliation of non-refundable fees recorded by agents.
- Sending customer notifications for approvals, rejections, and reminders.

C. Backend Services & Integrations

The backend orchestrates all workflows including:

- Customer KYC verification (Aadhaar/PAN).
- CIBIL and CRIF credit score integrations with secure logging.
- Fee recording and finance reconciliation tools.
- PDF generation for consent/sanction letters.
- Notification engine for SMS, Email, and WhatsApp messaging.
- Automated reminder scheduler (2–3 days before customer EMI dates).
- Secure document storage, audit logs, and role-based access.

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3. Project Scope

3.1 In-Scope Activities

A. Agent Mobile Application (Android & iOS)

Fully functional **Agent-only** application with:

1. Agent Authentication & Profile

- OTP login, biometric login support, admin approval for new agents.
- Profile management, device binding, logout, language preferences.

2. Customer Loan Application Creation

- Multi-step application form: customer basic info, contact, identity details, employment, income, loan request.
- Manual document uploads (Aadhaar, PAN, address proof, income proof, customer photos).
- NO OCR — all fields manually entered/uploaded.
- Draft creation & auto-save functionality.
- Application submission to generate **Applicant ID**.

3. KYC & Credit Verification

- Aadhaar/PAN verification through integrated KYC provider.
- CIBIL & CRIF credit score request initiation.
- Display bureau score summary and status updates.

4. Fee Collection (Offline)

- Record non-refundable application fee collected via **cash or branch deposit**.
- Upload receipt image, store metadata (amount, branch, date).
- Fee sync & finance reconciliation.

5. Approval & Consent Letter Flow

- After admin approval, agent receives automated **Loan Sanction Letter**.
- Notify customer via email/WhatsApp.

6. Post-Approval Customer Meeting

- Agent meets customer physically to:
    - Collect **bank account details** (account No., IFSC, account holder name).
    - Upload **signed disbursal form**.

  - Upload **SLA documentation**.
- Submit all final disbursal documents through the app.

7. EMI Calculator

- Built-in calculator for EMI, total payable, interest calculations.

8. Application Tracking & Notifications

- Real-time updates: Under Review, KYC Verified, Credit Completed, Approved, Rejected.
- In-app notifications for document requests and status changes.

9. Reminders

- Automated reminders to customers **2–3 days before due dates** (via SMS/WhatsApp/email).

B. Admin Web Portal

1. Admin Authentication & RBAC

- Secure login, role-based access (Admin, Underwriter, Finance, Operations).

2. Application Management

- View and manage all applications (search, sort, filter).
- Review customer details & documents.
- Manual verification of all uploaded documents (no OCR).
- Request additional documents or clarification from agents.

3. Underwriting & Credit Evaluation

- View CIBIL/CRIF results and scoring details.
- Add underwriting notes, approve or reject applications.
- Record rejection reasons and notify customer.

4. Loan Approval & Disbursal Preparation

- Convert Applicant ID → **Loan Account Number**.
- Trigger automated **Sanction Letter PDF** generation & delivery.
- Review bank details, disbursal form, SLA docs.
- Mark application as “Disbursal Ready”.

5. Finance & Fee Reconciliation

- Review offline payments with receipts.
- Reconcile agent-reported fees against branch/bank deposits.
- Mark payment status (Pending / Verified / Reconciled).

6. Reporting & Insights

- Application report (by agent, region, status).
- Fee collection report.
- Rejection analytics, TAT dashboards.

7. Audit & Compliance

- Full audit logs for all actions (approval, edits, login).
- Document version history.
- Secure data handling and retention policies.

8. Notification Management

- SMS/Email/WhatsApp delivery for Applicant ID, sanctions, rejections, reminders.
- Editable templates for communication.

C. Backend, APIs & Integrations

- Backend services for Agent App & Admin Portal.
- Integration with Aadhaar/PAN KYC provider.
- Integration with **CIBIL and CRIF**.
- Integration with SMS/Email/WhatsApp (notification provider).
- PDF generation engine (Sanction Letter).
- Automated reminder scheduler (EMI reminders).
- Encrypted document storage (Aadhaar, PAN, SLA docs, disbursal forms).
- API security, rate limiting, role-based access control.

D. Deployment & Support

- Cloud hosting setup for staging & production.
- CI/CD pipeline configuration.
- Testing (QA/UAT support).
- Admin & Agent training materials.
- Go-live assistance & warranty support as per agreement.

3.2 Out-of-Scope Items

The following are **NOT** included in Phase 1:

Loan Repayment / EMI Collection System

- No online EMI payment gateway.
- No UPI/NACH/eNACH auto-debit setup.
- No customer payment portal or ledger system.

Customer Mobile App

- Only agent-facing mobile app included.
- No direct customer login.

Automated Disbursal via Bank APIs

- Funds disbursal is offline and handled manually.
- No integration with bank disbursal APIs.

OCR / Automated Document Reading

- No automatic extraction of text from Aadhaar, PAN, bank statements, or income documents.

AI/ML Underwriting

- No machine learning scoring or predictive risk engine.

Collections & Recovery Module

- Only EMI reminders included.
- No delinquency tracking, field collections app, or payment follow-up module.

Multi-Tenant SaaS Version

- Single institution deployment only.

3.3 Assumptions

Business Assumptions

- Non-refundable application fee is collected **offline** (cash or branch deposit).
- Agents perform disbursal meetings physically with customers.
- Loan disbursal amounts handled outside the system (Phase 1).

Technical Assumptions

- Client will provide credentials & sandbox access for:
  - Aadhaar/PAN KYC provider
  - CIBIL & CRIF APIs
  - SMS/Email/WhatsApp services
- Document upload size limits and retention policies will be provided by the client.
- Mobile app supports latest OS + 2 previous versions.

Project Assumptions

- Client team will provide timely approvals and feedback for designs and UAT.
- SLA templates, sanction letter template, and disbursal form format will be provided by the client.
- Notifications (SMS/Email/WhatsApp) costs are borne by the client directly.

3.4 Constraints

Regulatory Constraints

- Compliance with RBI KYC norms and credit bureau usage rules.
- Aadhaar handling must follow UIDAI guidelines (masking, encryption, no unauthorized storage).

Technical Constraints

- App must support low bandwidth environments.
- Document uploads depend on network conditions.
- Bureau API response times are variable and outside system control.

Operational Constraints

- Offline fee collection depends on manual reconciliation by finance.
- Agent training is required to ensure proper data entry since there is **no OCR** automation.

4. Feature Listings

Agent Mobile App (Agent-only flows)

A. Authentication & Agent Management

- **Login / Registration**
  - Phone + OTP login; optional password fallback; device biometric option after first login.
  - Registration form (name, email, phone, agent ID/PAN, upload agent ID proof).
  - Admin approval required for new agents (Pending → Active).
- **Profile & Settings**
  - View/edit profile (editable fields require OTP for phone/email changes).
  - Device management (list devices, remote logout), language selection, notification preferences.

B. Dashboard

- KPIs: New Applications, Pending, Approved, Fees Collected.
- Quick Actions: New Application, EMI Calculator, Notifications, Upload Receipt.
- Recent activity and shortcuts to drafts and pending admin requests.

C. New Application Wizard (multi-step)

- **Step 1 — Customer Basic**
  - Fields: full name, DOB (age >= 18), gender.
  - Validations: name required, DOB not future.
- **Step 2 — Contact**
  - Primary phone (required, 10 digits), alternate phone, email.
  - Address: line1, line2, city, state, pincode.
  - Optional: geolocation capture (agent device GPS) for field verification.
  - Duplicate check: server-side soft check by phone/Aadhaar/PAN; show matches and allow admin override request.
- **Step 3 — Identity**
  - Fields: Aadhaar number (masked), PAN number (masked), ID type selection.
  - Uploads: scanned/photographed Aadhaar front/back, PAN image, selfie (if required), other ID proof.
  - Validation: format checks for Aadhaar/PAN; documents required per product rules.
  - Consent checkbox (see KYC & consent below).
- **Step 4 — Employment & Income**
  - Employment type (Salaried/Self-employed/Business/Agriculture), employer name, monthly/annual income, income proof upload (salary slip/bank statement/ITR).
  - Optional bank account placeholder (if agent collects bank details now, else will collect at consent meeting).
- **Step 5 — Loan Request**
  - Loan amount requested, tenure (months), purpose dropdown (personal/business/medical/education/other).
  - Product selection if applicable.

  - Shows **EMI estimate** (live) calculated on device (or via API).
- **Step 6 — Documents Upload**
  - Manual upload only: Aadhaar/PAN/Address proof/Income proof/Signatures/Dispersal forms later.
  - File size limits, allowed file types, and upload progress UI.
- **Step 7 — Review & Submit**
  - Review page with edit links for each section.
  - Checkbox: agent confirms documents are genuine and consent obtained.
  - Submit triggers business validation; on success returns **Applicant ID** (e.g., APP-YYYYMMDD-XXXX).
  - Show next steps: fee payment instructions, admin review ETA (SLA), and schedule for customer meeting to collect bank details & disbursal form.

D. KYC & Credit Flow

- **Consent Capture**
  - Capture explicit consent: show full consent text, record agent signature (finger), capture optional photo of signed physical consent form.
  - Persist immutable consent record (timestamp, agent id, geolocation).
- **KYC Verification**
  - Agent triggers KYC verification flows (Aadhaar eKYC via provider or manual verification flow depending on integration).
  - Store KYC status: Pending / Verified / Rejected.
- **Credit Bureau Checks**
  - Buttons to request CIBIL and/or CRIF check (configurable).
  - Asynchronous processing with progress UI (Pending → Completed → Failed).
  - Store raw report PDF and parsed numeric score; display score summary in-app.

E. Non-Refundable Fee Capture (Phase 1: offline)

- **Fee Requirement Notice**
  - On submit, app indicates required non-refundable application fee amount and payment channels (cash at partner branch / agent-collected cash).
- **Fee Recording**
  - Agent records payment: amount, collection method (cash / branch deposit), collector agent ID (auto), branch code (if branch), date/time.
  - Upload receipt image (photo of deposit slip or receipt).
  - App can generate a printable/WhatsApp-ready payment instruction slip showing Applicant ID, branch code, amount, and reference.
- **Offline Support**
  - Payment recording works offline (queued sync when online). Flag pending reconciliation until finance confirms.

F. Consent / Disbursal Meeting & Bank Details Capture

- **Consent Letter Generation**
  - After Approval (admin), agent receives generated Consent/Loan letter link and instruction to meet customer.
- **In-person Flow**
  - Agent meets customer with consent letter.
  - **Collect Bank Account Details**: account number, IFSC, account holder name, account type (savings/current).
  - **Collect Disbursal Form** (signed): agent uploads scanned disbursal form PDF/image.
  - **SLA Documentation**: capture SLA acknowledgement (customer signs or agent captures photo of signed SLA doc).
  - Agent uploads all gathered docs to application (these are required for final loan disbursal).
- **If Loan Rejected**
  - Agent receives rejection notification and in-app reason; customer must be notified (see Notifications).

G. Application Tracking & Messaging

- Status timeline: Draft → Submitted → Under Review → KYC Pending → Credit Check → Approved → Rejected → Disbursal Pending → Disbursed.
- In-app messaging/chat between agent and admin for clarifications; attachments allowed.
- Notifications (push/in-app) on status transitions and admin messages.

H. EMI Calculator

- Inputs: Principal, Annual Interest Rate, Tenure (months).
- Output: EMI, total interest, total payable, amortization schedule (monthly principal/interest).
- Button to **Apply** calculated EMI values to loan request form.

I. Reminders & Alerts

- Automatic reminders for customer: 2–3 days before due dates (configurable); agent gets in-app reminders of scheduled meetings or pending reconciliations.
- Alerts for pending fee reconciliation, missing docs, or SLA breaches.

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Agent Mobile App

Admin Web Application

A. Authentication & RBAC

- Secure login (SSO optional), MFA for admin accounts.
- Roles: Super Admin, Underwriting, Operations, Finance, Compliance, Agent Manager.
- Fine-grained permissions per module.

B. Dashboard & Monitoring

- KPIs: applications by status, avg. TAT, pending KYC, pending reconciliation, fees collected, agent performance.
- Alerts for SLA breaches, failed integrations, backlog.

C. Application Management

- **Application Queue**
  - Filterable, sortable list by agent, region, status, credit score, date.
- **Application Detail Page**
  - Full view of submitted data, uploaded docs, consent evidence, credit reports, payment receipts, SLA docs, messages, and audit log.
- **Document Verification**
  - Manual verification controls (Mark Verified / Reject / Request Reupload) for each uploaded document.
- **Underwriting**
  - View CIBIL/CRIF scores & flags; run configurable rule-based checks; add decision notes.
  - Approve / Reject actions with reason codes and optional conditions.
- **Loan Conversion**
  - On approval: convert Applicant ID → **Loan Account Number** (e.g., LN-YYYYMMDD-XXXX), set interest rate, finalize EMI schedule, mark for disbursal.
- **Consent / Letter Generation**
  - Trigger generation of consent/loan letter PDF and send via Email / WhatsApp to customer; mark letter as issued.
- **Disbursal Checklist**
  - Verify bank account info, signed disbursal form, SLA documentation and mark Disbursal Ready.
- **Rejection Workflow**
  - Send rejection notification with reason to customer (via agent channel, WhatsApp/SMS/email). Store rejection audit.

D. Payments & Finance

- **Fee Reconciliation UI**
  - List of unverified payments (receipt images), match to bank statements (uploaded CSV), mark reconciled and clear pending_reconciliation.
- **Manual Reconciliation Tools**
  - Allow finance to mark exceptions, request physical verification, or reverse entries.
- **Reports & Exports**
  - Export reports for fee collections, pending reconciliations, agent payouts, audit bundles.

E. Notifications & Templates

- Manage templates for SMS/Email/WhatsApp: submission confirmation (Applicant ID), approval (Loan Account), rejection, consent letter, payment instructions, reminders.
- Send one-off or automated notifications. Track delivery status.

F. Agent Management

- Approve new agent registrations; view activity logs; suspend/reactivate agents; assign regions/tiers.
- Agent performance dashboards (conversion rates, avg TAT, fees collected).

G. Audit, Compliance & Reporting

- Full immutable audit trails: who changed what and when.
- Compliance exports: application package (data + docs + consent + credit report).
- Data retention/purge controls and legal hold support.

Notifications & Customer Communication

- **Events to notify**
  - Application submitted (Applicant ID),
  - KYC completion/failure,
  - Credit check result summary,
  - Admin requests for documents,
  - Application approved → Loan Account Number,
  - Consent/loan letter issued,
  - Disbursal Ready / Disbursed,
  - Application rejected (with reason),
  - Reminders (2–3 days before EMI or payment due).
- **Channels**
  - In-app (agent), SMS, Email, WhatsApp (customer).
- **Templates**
  - Manageable templates with placeholders for names, Applicant ID, Loan Account, amounts, dates.

Payments & Reconciliation (Phase 1: Offline)

- Capture offline receipts with images and metadata.
- Generate payment instruction slips for customers to deposit cash at partner branch (shows branch code & Applicant ID).
- Finance imports bank statements and reconciles via UI (auto-match rules + manual override).
- Mark payments reconciled; only reconciled payments are considered cleared for underwriting / disbursal as per business rules.

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5. Non-Functional Requirements (NFRs)

1. Performance

- Application submission should complete within **≤ 3 seconds** on standard networks.
- Credit check responses handled asynchronously with updates within **≤ 30 seconds** (subject to bureau response).
- Dashboard pages load within **≤ 2 seconds** for typical data volumes.

2. Scalability

- System must support **multi-region growth**, increasing agents, and application volume without performance degradation.
- Architecture should scale horizontally for API load, document processing, and notification bursts.

3. Availability & Reliability

- Target **99.5% uptime** for core services.
- Background workers (credit checks, reminders) must retry automatically on transient failures.
- System must gracefully handle external API downtime (e.g., CIBIL, CRIF, eKYC).

4. Security

- End-to-end TLS encryption for all data in transit.
- Sensitive data (Aadhaar, PAN, documents) encrypted at rest using industry standards (AES-256).
- Role-Based Access Control (RBAC), MFA for admins, device-binding for agents.
- Compliance with **RBI**, **KYC**, **UIDAI**, and **credit bureau** guidelines.
- Full audit logs for all critical actions.

5. Data Privacy

- Mask Aadhaar/PAN in UI; store only hashed or tokenized identifiers where applicable.
- Strict retention policies for documents and consent records.
- Access to sensitive PII restricted based on user role.

6. Usability

- Simple, intuitive UX optimized for agents in field operations.
- Low-bandwidth mode and offline-first workflows for remote regions.
- Accessible UI with clear instructions, validation feedback, and consistent design.

7. Compatibility

- Agent mobile app to support Android (min OS version agreed) and iOS (min iOS version).
- Admin Portal optimized for modern browsers: Chrome, Edge, Safari.

8. Maintainability

- Modular code architecture enabling easy upgrades and addition of new features.
- API versioning and clear documentation.
- Logging, monitoring, and alerting integrated into infrastructure.

9. Disaster Recovery / Backup

- Daily automated backups with minimum **30-day retention**.
- DR strategy ensuring recovery within acceptable RPO/RTO limits.
- Redundancy for critical services and document storage.

10. Observability & Monitoring

- Real-time monitoring for API health, queue failures, bureau integrations, and worker jobs.
- Alerts for SLA breaches, errors, and unusual activity.

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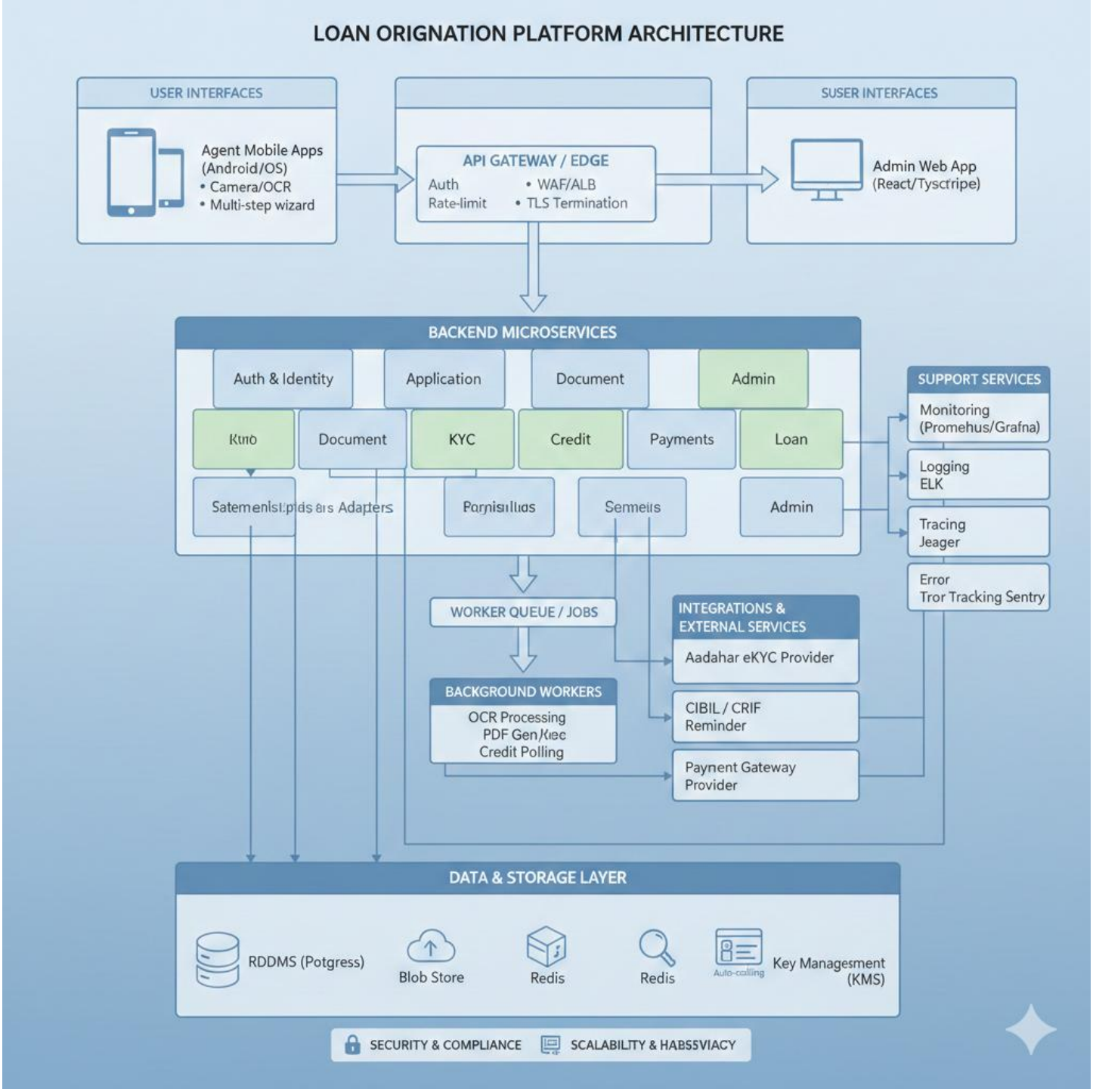
6. Technology Stack

Layer / Purpose	Recommended Tech	Rationale (short)
Mobile (Cross-platform)	React Native	Single codebase for Android+iOS; fast iteration; rich native module support (camera, biometrics).
Push Notifications	FCM (Android) + APNs (iOS)	Standard, reliable push delivery across platforms.
Mobile Crash / Analytics	Firebase Crashlytics + Sentry	Crash reporting + error/performance monitoring.
Admin Web Frontend	React (TypeScript) + Next.js	Fast, component-driven, great DX with TypeScript for safety.
UI / Styling	Tailwind CSS + shadcn/ui (or Chakra)	Modern, accessible UI; aligns with proposal styling guidance.
State / Data Fetch	Redux Toolkit	Server-state caching + lightweight local state.
Backend Framework	Node.js (TypeScript)	Modular, TypeScript-first, good for microservices.
API Style	REST (JSON) — versioned (/api/v1)	Mobile friendly; predictable & simple; idempotency support.
Primary Database	MongoDB	ACID, relational modeling + JSONB for flexible fields.
Cache / Broker	Redis	Caching, locks, job metadata, and pub/sub for workers.
Background Jobs	BullMQ (Redis-backed)	Robust job processing for credit polling, PDF gen.
Document Storage	AWS S3 / Azure Blob / GCS	Durable, scalable object storage with signed URLs & lifecycle rules.
Search & Duplicate Detection	Elasticsearch / OpenSearch	Fuzzy search and fast duplicate detection across records.

Credit / KYC Integrations	Provider adapters (CIBIL, CRIF, eKYC aggregator)	Encapsulate bureau-specific logic, retries, logging.
Notifications (SMS/WhatsApp/Email)	Twilio / Gupshup / MessageBird + SendGrid	Multi-channel delivery; WhatsApp Business API support.
PDF Generation	Puppeteer (HTML→PDF) or PDFKit	Template-driven PDF generation for agreements & receipts.
Containerization / Orchestration	Docker + Kubernetes (EKS / GKE / AKS)	Scalable deployments, rolling updates, auto-scaling.
CI / CD	GitHub Actions / GitLab CI	Automated build, test, and deploy pipelines.
Infrastructure as Code	Terraform + Helm (K8s charts)	Reproducible infra provisioning and deployments.
Secrets / KMS	AWS KMS / HashiCorp Vault / Azure Key Vault	Secure key & secret management with rotation.
API Gateway / Edge	AWS API Gateway / Kong / NGINX + WAF	Central auth, throttling, TLS termination, security policies.
Monitoring & Logging	Prometheus + Grafana; ELK/EFK or Datadog; Sentry	Metrics, dashboards, traces, centralized logs & error tracking.
Load / Security Testing	k6; OWASP ZAP; Snyk / Dependabot	Performance & vulnerability scanning.
BI / Reporting	Fivetran / Airbyte → Snowflake / BigQuery; Metabase / Looker	Data pipelines for analytics & reporting.
Dev Tools / Local Stack	Docker Compose / LocalStack (for AWS emulation)	Local developer productivity and integration testing.
Compliance / Audit	Immutable audit logs (ELK), DLP scripts	Support for regulatory audits & consent retention.


LOAN ORIGNATION PLATFORM ARCHITECTURE
USER INTERFACES
Agent Mobile Apps (Android/OS)
• Camera/OCR
• Multi-step wizard
API GATEWAY / EDGE
Auth
Rate-limit
• WAF/ALB
• TLS Termination
SUSER INTERFACES
Admin Web App (React/Tysctripe)
BACKEND MICROSERVICES
Auth & Identity
Application
Document
Admin
Document
KYC
Credit
Payments
Loan
Admin
SUPPORT SERVICES
Monitoring (Promehus/Grafna)
Logging ELK
Tracing Jeager
Error Tror Tracking Sentry
WORKER QUEUE / JOBS
BACKGROUND WORKERS
OCR Processing
PDF Gen/Kiec
Credit Polling
INTEGRATIONS & EXTERNAL SERVICES
Aadahar eKYC Provider
CIBIL / CRIF Reminder
Paynent Gateway Provider
DATA & STORAGE LAYER
RDDMS (Potgress)
Blob Store
Redis
Redis
Key Managesment (KMS)
SECURITY & COMPLIANCE
SCALABILITY & HABSSVIACY

7. Receivables from Client

The client will take care of the following:

1. **Developer Accounts:**
    1. **Apple Developer Account:** $99/year (App store)
    2. **Android Developer Account:** $25 one-time fee (Playstore)
2. **Third-party APIs:** The client will provide necessary APIs and subscriptions (e.g., Payment Gateway, SMS Gateway, Cloudinary, etc.).
3. **Application Name, Logo and Content:** The client will provide the app's name, logo, content. If unavailable, we will provide **at an additional cost.**
4. **Firebase Account:** Required for push notifications.
5. **SMS Gateway:** Client to provide an SMS gateway.
6. **Cloudinary/Image Server:** For image storage.
7. **Additional Features at additional cost:** The client will notify us if any additional features are needed outside of the current scope.
8. **Design Preferences:** The client should provide recommendations for the application theme and color scheme.

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8. Deliverables to Client

The following key deliverables will be provided to the client as part of this project:

1. Agent Mobile Applications (Android & iOS)

- Fully functional agent-only mobile app
- Customer onboarding & loan application workflow
- KYC, credit check, fee recording, EMI calculator
- Offline support + sync functionality
- App build files (APK/IPA/TestFlight deployment)

2. Admin Web Application

- Complete admin portal for operations, underwriting, finance & compliance
- Application review, verification & approval workflows
- Loan ID generation & document management
- Fee reconciliation dashboards & reporting tools
- Role-based access control setup

3. Backend Services & Integrations

- API layer for mobile & web
- Integration with Aadhaar eKYC, PAN verification, CIBIL/CRIF credit bureaus
- Notification engine for SMS/Email/WhatsApp
- PDF generation engine (loan agreements, consents, receipts)

4. Infrastructure Setup

- Cloud deployment (staging & production)
- Secure storage, authentication, logging, audit trails
- Database & backup configuration

5. Documentation

- Technical documentation (API specs, architecture diagrams)
- User manuals for Agents & Admins
- Setup guides and configuration notes

6. Testing & Quality Assurance

- Test cases, UAT support, QA results
- Bug fixes and stabilization prior to go-live

7. Go-Live & Post-Launch Support

- Production deployment
- Warranty support (as per agreement)
- Training sessions for admin/agent teams

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9. Acceptance Criteria

The project will be considered successfully delivered when the following acceptance criteria are met:

1. Agent Mobile App Functionality

- Agents can register, log in, and operate the app smoothly (online/offline).
- Agents can complete and submit loan applications with all required fields, documents, and KYC details.
- Aadhaar eKYC, PAN verification, and Credit Bureau checks (CIBIL/CRIF) function reliably.
- Application submission generates an **Applicant ID** and syncs to the admin portal.
- EMI calculator, fee recording, and document uploads operate with expected accuracy.

2. Admin Web Portal Functionality

- Admins can view, verify, approve, or reject applications end-to-end.
- Loan approval successfully converts **Applicant ID → Loan ID** and auto-generates loan documents.
- Payment/fee records are visible and reconcilable.
- Notification templates (SMS/Email/WhatsApp) send correctly for submissions, approvals, and reminders.
- All admin modules (agents, applications, documents, reports) function as per specification.

3. Integrations & Backend

- Aadhaar eKYC, PAN verification, CIBIL & CRIF integrations return valid results with audit logs.
- Backend APIs operate within defined performance limits and are secure (encrypted, authenticated).
- PDF generation (consent, loan agreement, receipts) is accurate and delivered to customer channels.

4. Quality, Security & Compliance

- All critical paths pass UAT test cases with zero high-severity defects.
- No PII leakage; Aadhaar/PAN masking and encryption meet compliance rules.
- Audit log captures all major actions (agent submissions, admin approvals, document changes).

5. Deployment & Support

- Staging and Production environments are deployed, tested, and stable.
- Client receives required documentation, training, and knowledge transfer.
- Warranty/support period begins after client’s formal sign-off.

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10. Project Plan & Milestones

Phase / Milestone	Activities Included	Expected Duration	Deliverables
1. Project Kickoff & Requirement Finalization	Stakeholder meetings, requirement clarification, scope confirmation, technical discussion, integration requirement collection		Final BRD, SRS, Project Charter
2. UI/UX Design	Wireframes for Agent App & Admin Web, user flows, high-fidelity screens, design approval cycle		Approved UI/UX Screens & Clickable Prototype
3. Backend Architecture & Setup	Database design, API structure, KYC/credit integration planning, environment setup (dev/staging)		Architecture Document, Dev Environment Ready
4. Agent Mobile App – Core Module Development	Login, registration, dashboard, draft application flow, offline sync structure		Mobile App Core Module (Alpha Build)
5. Agent Mobile App – Application Workflow Development	Customer onboarding steps, document upload, consent, KYC, credit check, fee recording, EMI calculator		End-to-End Agent Application Workflow (Beta Build)
6. Agent Web Application	Login, registration, dashboard, draft application flow, offline sync structure		Web Application
7. Admin Web Portal – Phase 1 Development	Login/RBAC, dashboard, application queue, detailed view, document verification		Admin Portal Core (Beta Build)
8. Admin Web Portal – Phase 2 Development	Underwriting tools, approval workflow, Applicant ID → Loan ID conversion, fee reconciliation		Fully Functional Admin Portal
9. Integrations Development	Aadhaar eKYC, PAN verification, CIBIL/CRIF APIs, SMS/Email/WhatsApp notifications, PDF generator		Integrated & Tested External Services

10. QA, Testing & Bug Fixing	Functional testing for agent & admin, integration tests, load tests, security checks, UAT support	UAT-ready builds & Test Reports
11. User Acceptance Testing (UAT)	Client-side testing, feedback cycles, minor change requests, functional validation	UAT Sign-off
12. Deployment (Staging & Production)	Staging deployment, final checks, production deployment, data initialization	Live System (Agent App + Admin Portal)
13. Training & Handover	Admin training, agent training materials, knowledge transfer, documentation delivery	Training Sessions & Documentation

Total Estimated Timeline:

11. Cost Analysis

Cost Category	Estimated Cost (₹ / $)
1. UI/UX Design	₹
2. Agent Mobile App Development (Android + iOS)	₹
3. Agent Web Application	
4. Admin Web Portal Development	₹
5. Backend API & Database Development	₹
6. Third-Party Integrations	₹
7. PDF Generation System	₹
8. Notification Engine	₹
9. DevOps, Cloud Setup & Deployment	₹
10. QA & Testing	₹
11. Project Management	₹
12. Training & Handover	₹

Total Estimated Project Cost: ₹

Payment Terms

The payment will be divided into the following milestones:

1. **30% upfront**: Before starting the project.
2. **20% on Design approval**: After design finalization.
3. **35% on mid of development**: Post-integration of backend and frontend.
4. **15% on delivery**: On final testing and go-live.

Important Notes:

- The exact speed of design& development will be affected by client turnaround times, for example responding to designs, how much feedback is given etc.
- Services are supplied according to our terms and conditions: The above estimate is based ONLY on the features which we listed in this document.
- If you need any other feature (NOT listed in this doc) then the Time Estimate and cost may ramp up. It may also vary after sharing your response on the queries being asked in this document.
- Any delay in reviewing and approving work, providing APIs, or responding to queries will result in pushing the project deadline and extending the overall project timeline.

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12. Change Request

Any modifications to the approved scope, features, timelines, or deliverables will follow a formal **Change Request (CR)** process. A CR will be raised when new requirements emerge, existing features need enhancement, or third-party dependencies change. Each Change Request will include:

- **Description of requested change**
- **Reason for change**
- **Impact analysis** on cost, timeline, and resources
- **Updated effort estimate**
- **Approval from client stakeholders**

No change will be implemented without **written confirmation** from the client. Approved changes will be added as separate deliverables or milestones, and billed accordingly.

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13. Maintenance & Support Plan

We provide **3 months of complimentary maintenance and support** post-launch. This includes bug fixes, performance optimizations, and minor feature adjustments. Any additional support or major feature updates will be available at an additional cost based on the scope of the request.

Our maintenance and support plan ensures smooth operation, system stability, and ongoing updates to meet evolving business needs. It includes:

- Bug Fixing: Rapid resolution of critical and non-critical issues.
- Performance Monitoring: Real-time system health checks and optimization.
- Security Updates: Regular patches to address vulnerabilities and enhance system safety.
- Feature Enhancements: Periodic updates and new feature rollouts based on client feedback.
- 24/7 Technical Support: Dedicated team available for troubleshooting and assistance.

Support Tier	Services Included	Monthly Cost (₹)
Basic Support	Bug fixes, security updates, and performance monitoring	
Standard Support	Basic Support + minor feature enhancements	
Premium Support	Standard Support + 24/7 technical support and priority fixes	

Conclusion

The proposed Loan Application Platform — consisting of a field-focused Agent Mobile Application and a robust Admin Web Portal — is designed to digitally transform the client's lending operations. By streamlining customer onboarding, KYC verification, credit evaluation, fee recording, and loan approval processes, the solution empowers the organization to scale efficiently across Tier 1, Tier 2, and small-city markets.

This platform not only reduces operational bottlenecks and manual errors but also enhances compliance, audit readiness, transparency, and customer satisfaction. With strong foundations in security, scalability, and modern technology, the system positions the client as a forward-thinking, digitally enabled lending institution.

Through clear workflows, role-based controls, automation, and seamless integrations, the platform delivers faster turnaround times and higher-quality loan decisions while strengthening agent productivity and overall operational efficiency. The project sets the stage for future expansion — including online repayment, customer self-service, automated underwriting, and advanced analytics.

By adopting this solution, the client invests in long-term operational resilience, competitive differentiation, and improved financial performance. The result is a modern, scalable, and customer-centric lending ecosystem ready to support growth today and innovation tomorrow.